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Reconstructing a quantum state with a variational autoencoder

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Quantum state tomography (QST) is an important and challenging task in the field of quantum information, which has attracted a lot of attentions in recent years. Machine learning models can provide a classical representation of the quantum state after trained on the measurement outcomes, which are part of effective techniques to solve QST problem. In this work, we use a variational autoencoder (VAE) to learn the measurement distribution of two quantum states generated by MPS circuits. We first consider the Greenberger–Horne–Zeilinger (GHZ) state which can be generated by a simple MPS circuit. Simulation results show that a VAE can reconstruct 3- to 8-qubit GHZ states with a high fidelity, i.e., 0.99, and is robust to depolarizing noise. The minimum number (N_s^*) of training samples required to reconstruct the GHZ state up to 0.99 fidelity scales approximately linearly with the number of qubits (N). However, for the quantum state generated by a complex MPS circuit, N_s^* increases exponentially with N, especially for the quantum state with high entanglement entropy.

Keywords: Quantum state tomography; quantum machine learning; variational autoencoder.

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1. Introduction

Quantum state tomography (QST) has received a lot of attention in recent years, which aims to get a complete description of a quantum state by measurements and then reconstruct the density matrix ρ. However, QST is impracticable in reality, due to the fact that the dimension of the reconstructed density matrix increases exponentially with the number of qubits, i.e. the so-called "curse of dimensionality". There are several methods to solve the QST problem, including general maximum likelihood estimation (MLE),[1,2] Bayesian tomography,[3,4] matrix product state tomography[5,6] and neural network tomography.[7–10] Machine learning technique is an effective way to solve the QST problem. A machine learning model can be a classical representation of a quantum state after trained on measurement outcomes of the quantum state. Reference 7 reconstructed the quantum state with a recurrent neural network (RNN) and got a high-fidelity reconstruction. Reference 10 proposed an attention-based state tomography, which used a transformer model[11] to reconstruct the quantum state, and their simulation result showed that the transformer could reconstruct the Greenberger–Horne–Zeilinger (GHZ) state even with noisy samples.

In this paper, we use a variational autoencoder (VAE)[12] to reconstruct quantum states. VAE is a simple and powerful generative model which can learn the probability distribution from a set of samples. More specifically, we use a VAE to learn the quantum state from a set of measurement outcomes. A VAE is first trained on a set of positive-operator valued measure (POVM) outcomes of the quantum state in order to learn its abstract representation, and then the trained VAE can generate a number of samples, which approximate the distribution of POVM outcomes of the quantum state. The density matrix of the quantum sate can be reconstructed based on the generated samples by the trained VAE. In this paper, we consider two target states generated by a simple and a complex MPS circuits, i.e. the GHZ state and the state generated by the MPS circuit in Ref. 13. Simulation results show that VAE can learn the true probability distribution from a set of quantum state measurements with high fidelity. Due to the simple structure, the number of measurements required to reconstruct the GHZ state is linearly related to the number of qubits. However, for the quantum state generated by a more complex MPS circuit, the number of required measurements increases exponentially with the number of qubits.

Similar work is found in Ref. 14. Instead of QST, Rocchetto *et al.* used a VAE to encode the probability distribution of quantum states, which cannot reconstruct the full set of amplitudes of the quantum state. Besides, they focused on the compression rate and how the depth of VAE affects the reconstruction accuracy. However, we consider to solve the QST problem by a VAE, which is trained on a number of measurements on the quantum state using informationally complete (IC) POVM. We focus on how the number of training samples required to reconstruct a quantum state scales with the number of qubits. Reference 15 showed that the number of measurements to determine the fidelity of the GHZ state increased linearly with the number of qubits. Different from their work, we aim to reconstruct a GHZ state according to a number of measurement outcomes of the GHZ state and analyze how the number of measurements required to reconstruct the GHZ state scales with the number of qubits. Ahmed *et al.* proposed a reconstruction of optical quantum states using conditional generative adversarial networks (CGAN).[16] The input data are the measurement statistics and the operators that are measured. They estimated the full density matrix of the target state by the outputs of an intermediate DensityMatrix layer. In the proposed method, we train a VAE with multiple measurement outcomes of the quantum state. The input data to train the model are a number of measurement outcomes of the quantum state, rather than a statistical result on the entire dataset.

The rest of this paper is organized as follows. Section 2 describes how to reconstruct a quantum state with a VAE. Section 3 shows numerical results of the proposed method on the reconstruction of the quantum states generated by MPS circuits, and analyzes how the minimum number of measurements required by VAE scales with the number of qubits. The conclusion is drawn in Sec. 4.

2. Methods

2.1. *Reconstructing quantum state with VAE*

A VAE is a powerful generative model, which poses little difficulty to generate natural images,[17,18] speech[19] and text.[20] In this paper, we use a VAE to learn from measurement outcomes of the quantum state and then reconstruct the approximate density matrix with the generated samples by the decoder of the trained VAE. Our method consists of two steps, i.e. training process and generating process.

The flowchart of training a VAE for QST is illustrated in Fig. 1. We conduct N_s measurements on the quantum state using informationally complete (IC) POVMs and each measurement outcome is recorded as a training sample of the training set. Specifically, we use the 4-outcome POVM on ideal quantum states where there are

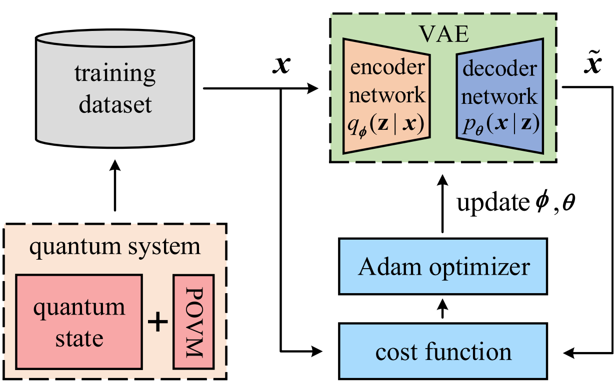


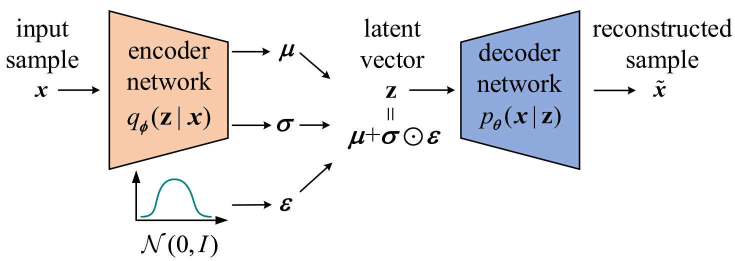
Fig. 1. The flowchart of training a VAE for QST. ϕ and θ are the parameters of the encoder and decoder in VAE.

four possible outcomes (denoted by 0, 1, 2 and 3) for each qubit. We use a one-hot vector (1×4) to distinguish four possible outcomes. Thus, each training sample can be denoted by a $4N$-dimensional vector where N is the number of qubits. In each training iteration, we randomly draw a batch of samples from the training set and update the parameters of VAE to minimize the cost function with respect to these samples. The iteration repeats until the cost function converges.

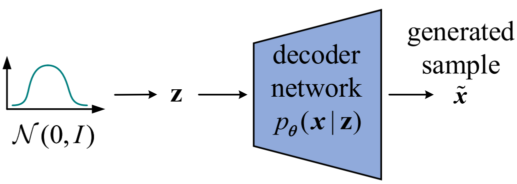
VAE consists of an encoder neural network q_{ϕ} and a decoder neural network p_{θ}, as shown in Fig. 2. During the training process, the encoder takes the training sample $\boldsymbol{x}$ as input, and outputs a mean vector $\boldsymbol{\mu}$ and a standard deviation vector $\boldsymbol{\sigma}$. According to the reparameterization trick in Ref. 12, the latent vector can be obtained by $\boldsymbol{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\varepsilon}$, which captures the principal characteristics of the samples. $\boldsymbol{\varepsilon}$ is a random vector sampled from a standard normal distribution $\mathcal{N}(0, I)$ and $\odot$ is the element-wise multiplication. Then the decoder p_{θ} takes the latent vector $\boldsymbol{z}$ as input and generates the reconstructed sample $\tilde{\boldsymbol{x}}$. The training of VAE aims to maximize the Evidence Lower Bound (ELBO) which is equal to minimizing the loss function

$$J(\theta, \boldsymbol{\phi}, \boldsymbol{x}) = -\mathbb{E}_{\boldsymbol{z}\sim q_{\phi}(\boldsymbol{z}|\boldsymbol{x})} \log p_{\theta}(\boldsymbol{x}|\boldsymbol{z}) + D_{KL}(q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x}) \| p_{\theta}(\boldsymbol{z})), \tag{1}$$

where $\boldsymbol{\phi}$ and θ are the parameters of the encoder and decoder networks, respectively. The first term is reconstruction loss, which identifies the difference between the reconstructed sample and the input sample. The second term is regularization loss, which calculates the Kullback–Leibler (KL) divergence between the distribution of encoder $q_{\boldsymbol{\phi}}(\boldsymbol{z}|\boldsymbol{x})$ and the prior distribution $p(\boldsymbol{x})$ (i.e. a standard Gaussian distribution $\mathcal{N}(0, I)$). It is a regularization term that acts on the latent space, which can avoid



(a)



(b)

Fig. 2. The architecture of variational autoencoder. (a) The training process and (b) the generating process.

overfitting. The KL divergence between two distributions P_1 and P_2 is defined by

$$D_{\mathrm{K\,L}}(P_1 \| P_2) = -\sum_{\boldsymbol{x}} P_1(\boldsymbol{x}) \log \frac{P_2(\boldsymbol{x})}{P_1(\boldsymbol{x})}. \tag{2}$$

After the training process, the trained decoder can generate an arbitrary number of samples, which approximate the distribution of the training samples generated by POVM outcomes of the quantum state, with respect to the input of random vectors sampled from a standard normal distribution $\mathcal{N}(0, I)$, as described in Fig. 2(b). In the generating process, an output of the trained VAE is one of the possible measurement outcomes, which is denoted by a $4N$-dimensional vector. The probability distribution (P_{vae}) of the trained VAE can be estimated by the generated N_e samples. The classical fidelity between the reconstructed distribution (P_{vae}) and the exact one (P) can be calculated by

$$F_c(P_{\mathrm{vae}}, P) = \sum_{\boldsymbol{x}} \sqrt{P_{\mathrm{vae}}(\boldsymbol{x}) P(\boldsymbol{x})}. \tag{3}$$

As the POVM we used is IC, the density matrix can be unambiguously inferred from the probability distribution of measurement outcomes. According to Born's rule, the probability distribution $\mathbf{P} = \{P(\mathbf{a})\}_{\mathbf{a}}$ of measurement outcomes $\mathbf{a} = (a_1, a_2, \ldots, a_N)$ of an N-qubit quantum state ϱ is given by

$$P(\mathbf{a}) = \mathrm{Tr}[M^{(\mathbf{a})} \varrho], \tag{4}$$

where $M^{(\mathbf{a})} = \{M^{(a_1)} \otimes M^{(a_2)} \otimes \cdots \otimes M^{(a_N)}\}$ and $a_i \in \{0, 1, \ldots, m-1\}$ is the measurement outcome of the ith qubit. The overlap matrix $T_{\mathbf{a},\mathbf{a}'} = \mathrm{Tr}[M^{(\mathbf{a})} M^{(\mathbf{a}')}]$ is invertible if the measurement is IC. Therefore, Eq. (4) is invertible and the quantum state ϱ can be calculated by

$$\varrho = \sum_{\mathbf{a},\mathbf{a}'} P(\mathbf{a}) T^{-1}_{\mathbf{a},\mathbf{a}} M^{(\mathbf{a}')} = \mathbb{E}_{\mathbf{a}\sim\mathbf{P}} \left(\sum_{\mathbf{a}'} T^{-1}_{\mathbf{a},\mathbf{a}'} M^{(\mathbf{a}')} \right). \tag{5}$$

Thus, we can reconstruct the quantum state ϱ based on the reconstructed distribution (P_{vae}).

2.2. *Quantum states to be reconstructed*

The first quantum state we reconstruct is the GHZ state, which is a pure state commonly used in theoretical studies and specified by

$$|\Psi_{\mathrm{GHZ}}\rangle = \frac{1}{\sqrt{2}} (|0\rangle^{\otimes N} + |1\rangle^{\otimes N}), \tag{6}$$

where N is the number of qubits.

Another target state we consider is the quantum state generated by an MPS circuit proposed in Ref. 13. MPS circuits are especially advantageous for machine learning tasks and have an inherent resilience to noise. MPS circuits have been widely used in quantum machine learning models including quantum discriminative models and generative models. We use an MPS circuit with virtual qubits $V = 1$ and bond dimension $D = 2$. We define a 2-qubit unitary block $U = e^{iH}$ where H is a Hermitian matrix. A general N-qubit unitary can be expressed as follows:

$$U = \exp\left[i\left(\sum_{j_1,\ldots,j_N=0,\ldots,0}^{3,\ldots,3} \alpha_{j_1,\ldots,j_N}(\sigma_{j_1} \otimes \ldots \otimes \sigma_{j_N})\right)\right]. \tag{7}$$

H can be expressed as a linear combination of tensor products of Pauli and identity matrix. σ_0 is the 2×2 identity matrix and σ_i is the Pauli matrix where $i \in \{1, 2, 3\}$. Different unitary blocks U can be generated by various scalars α. For an N-qubit MPS circuit, we sequentially add a unitary block on the nth and $(n+1)$th qubits where $n = 1, 2, \ldots, N - 1$, as described in Fig. 3. It should be noted that a GHZ state has permutational symmetry and can be generated by a simple MPS circuit. A 3-qubit GHZ state can be generated from a $|000\rangle$ state by putting a Hadamard gate on the first qubit and a subsequent controlled-NOT gate between the first and the second qubits and another controlled-NOT gate between the second and the third qubits.

2.3. *Measurements*

2.3.1. *IC POVMs*

To obtain complete information about the quantum state from measurement, we consider the tetrahedral POVMs:

$$\mathbf{M}_{\text{tetra}} = \left\{M^{(\alpha)} = \frac{1}{4}(\mathbf{I} + \boldsymbol{s}^{(\alpha)} \cdot \boldsymbol{\sigma})\right\}_{\alpha\in(0,1,2,3)}, \tag{8}$$

where $\boldsymbol{\sigma} = (\boldsymbol{\sigma}_x, \boldsymbol{\sigma}_y, \boldsymbol{\sigma}_z)$. POVM outcomes correspond to sub-normalized rank-1 projectors along the directions $s^{(0)} = (0, 0, 1)$, $s^{(1)} = (\frac{2\sqrt{2}}{3}, 0, -\frac{1}{3})$, $s^{(2)} = (-\frac{\sqrt{2}}{3}, \sqrt{\frac{2}{3}}, -\frac{1}{3})$ and $s^{(3)} = (-\frac{\sqrt{2}}{3}, -\sqrt{\frac{2}{3}}, -\frac{1}{3})$.

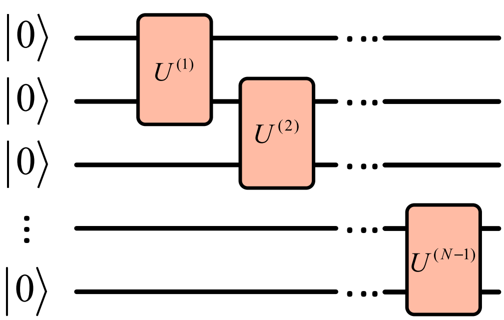


Fig. 3. An N-qubit MPS circuit.

2.3.2. *The Meyer–Wallach (MW) measure*

In this paper, we use the Meyer–Wallach (MW) measure[21] to calculate the global entanglement of the quantum state. MW measure is a function on an N-qubit pure state $|\psi\rangle \in (\mathbb{C}^2)^{\otimes N}$ defined as follows:

$$Q(|\psi\rangle) = \frac{4}{N} \sum_{k=1}^{N} D(|\tilde{u}^k\rangle, |\tilde{v}^k\rangle), \tag{9}$$

where $|\tilde{u}^k\rangle$ and $|\tilde{v}^k\rangle$ are non-normalized vectors. $|\psi\rangle$ can be projected onto the kth qubit subspace:

$$|\psi\rangle = |0\rangle_k \otimes |\tilde{u}^k\rangle + |1\rangle_k \otimes |\tilde{v}^k\rangle. \tag{10}$$

D calculates the distance of two vectors $|\tilde{u}^k\rangle$ and $|\tilde{v}^k\rangle$:

$$D(|\tilde{u}^k\rangle, |\tilde{v}^k\rangle) = \sum_{i<j} |\tilde{u}_i^k \tilde{v}_j^k - \tilde{u}_j^k \tilde{v}_i^k|^2. \tag{11}$$

3. Numerical Results

3.1. *Settings*

We use the PaddlePaddle framework[22] to train the VAE model. Leaky ReLU and sigmoid are used as the activation functions of the hidden layer and output layer, respectively. A warm-up schedule[23] is used on the regularization loss, and the weight of the second term D_{KL} in Eq. (1) is increased linearly from 0 to 0.85 during training. We use a stochastic gradient optimizer, i.e. Adaptive Moment Estimation (Adam),[24] to update the parameters of the VAE model with the learning rate 0.001. The number of iterations for VAE training is set to 10^6. In the training process, we conduct N_s measurements of the target quantum state and get the training set. A batch of samples are randomly selected from the training set to optimize the parameters of VAE at each iteration. In the generating process, we generate N_e samples from the VAE decoder to estimate the fidelity between the reconstructed distribution and the exact one.

We use a four-layer neural network as the encoder and decoder of the VAE model, which has one input layer, two hidden layers and one output layer. The number of neurons in the input layer of the encoder is equal to $4N$, and the number of neurons in the hidden and output layers varies with N. Table 1 shows the parameter setting in the simulation. We use a string to represent the layout of the encoder. For instance, the layout 12-96-96-16 means that the encoder has 12 neurons in the input layer, 96 hidden neurons in each hidden layer and 16 neurons in the output layer. The layout of the decoder is symmetrical to the encoder.

Table 1. Parameters setting in the simulations.

N	Layout	Batch size	N_e
3	12-96-96-16	100	$20,000$
4	16-128-128-32	200	$100,000$
5	20-224-224-64	500	$4^N * 500$
6	24-1280-1280-256	600	$4^N * 500$
7	28-1504-1504-512	800	$4^N * 500$
8	32-2560-2560-512	1000	$4^N * 500$

3.2. *Reconstructing a GHZ state*

We first investigate the number of training samples required to achieve a high reconstructed fidelity for GHZ states with different qubits. Figure 4 shows the reconstructed fidelity by the VAEs with the increasing number N_s of training samples. Each point is an average from 10 independent simulations and the error bars represent one standard deviation statistical uncertainty calculated over 10 simulations. As expected, the fidelity F_c increases with the number of training samples N_s. More POVM outcomes carry more information about the quantum state, which enable the VAE to represent the density matrix more accurately. With the increasing number of qubits, more training samples are needed to achieve a high reconstructed fidelity. When N_s is equal to 10^5, the reconstructed fidelities for all the quantum states are higher than 0.99, which demonstrates the high repressive power of VAE model. We also analyze the minimum number of training samples N_s^* needed to reach a 0.99 fidelity for GHZ states with different numbers of qubits, and find that N_s^* grows linearly with the number of qubits as shown in Fig. 5.

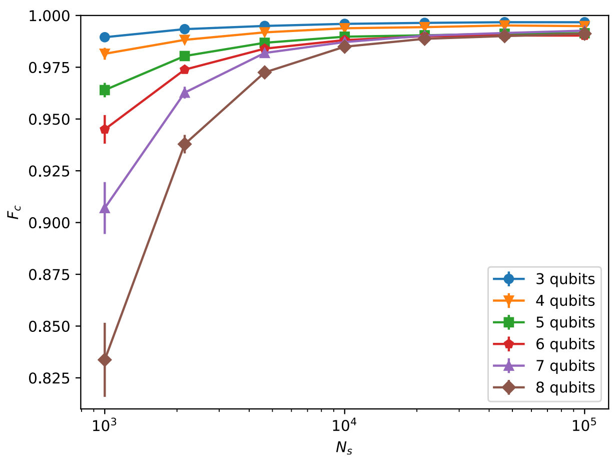


Fig. 4. The reconstructed fidelity for GHZ states with 3 to 8 qubits with the increasing number of training samples. Each point is an average of the results from 10 independent simulations and the error bars represent the standard deviation.

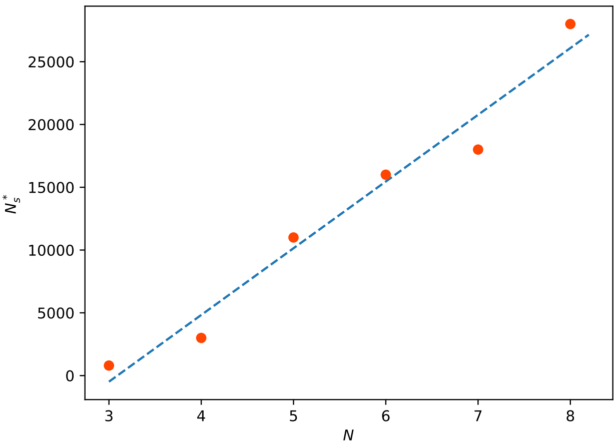


Fig. 5. The minimum number of training samples needed to achieve 0.99 fidelity for GHZ states.

In the NISQ-era, quantum devices are noisy and error prone. It is important to study the influence of noise on the reconstruction effect. We independently impose a depolarizing noise on each qubit, all information of the qubit is lost with a probability p. Thus, the depolarizing noise leads to the state of the form $(1-p) * \rho_i + p * \text{Identity}/2$, where ρ_i is the density matrix of the ith qubit. The training samples of VAE are the POVM outcomes of the GHZ state with depolarizing noise. The fidelity is calculated between the ideal GHZ state and the reconstructed one. The simulation results are shown in Fig. 6, which indicates that VAE can reconstruct the density matrix of the quantum states with depolarizing noise with a high fidelity. The fidelity F_c decreases linearly with the increase of the noise rate p.

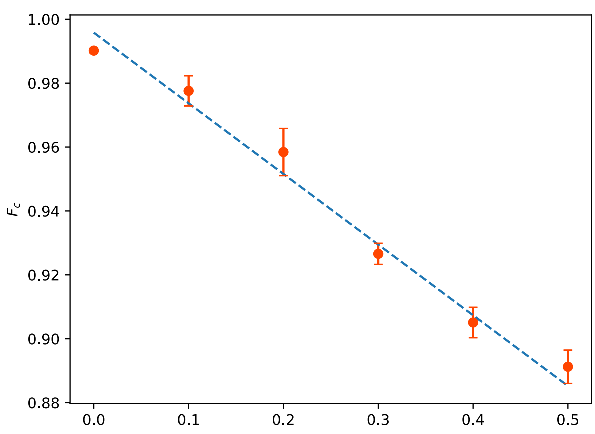


Fig. 6. The fidelity F_c of the reconstructed density matrix by VAE for GHZ states with different levels of noises. We set $N = 5$, $N_s = 11,000$. Each point is an average of the fidelities in 10 independent runs.

When $p = 0.5$, VAE can still reconstruct the quantum state with a fidelity of 0.891, which demonstrates the robustness of VAE model.

3.3. *Reconstructing the quantum state generated by a more complex MPS circuit*

The simulation result shows that VAE has a good performance on reconstructing the GHZ state, which is a simple state generated by a special MPS circuit. However, most of the quantum states are not as simple and relatively regular as the GHZ state. Thus, we also investigate the performance of VAE on reconstructing a quantum state generated by a more complex MPS circuit as shown in Fig. 3. Similar to the GHZ state, we generate the training samples by POVMs for the quantum state generated by an MPS circuit. We found that it was difficult for the reconstruction of this quantum state by VAE to reach a fidelity 0.99, especially for the quantum states with large system sizes, which shows that the quantum state generated by a complex MPS circuit is more difficult to reconstruct than the GHZ state. We set the target fidelity of reconstruction to 0.95, and estimate the minimum number of training samples required to train the VAE. Different from the GHZ state, the minimum number of training samples required to reach 0.95 fidelity increases exponentially with the number of qubits, as shown in Fig. 7. The reason might be that the probability distribution of the measurement outcomes of the quantum state generated by this MPS circuit is more complex and irregular than that of the GHZ state. We also analyze the reconstructions on the quantum states with different entanglement entropies, i.e. $Q(|\psi\rangle) = 0.2, 0.5, 0.8$. The entanglement entropy of the quantum state generated by the MPS circuit is calculated by the MW measure according to Eq. (9). For the quantum state with larger entanglement entropy, more training samples are

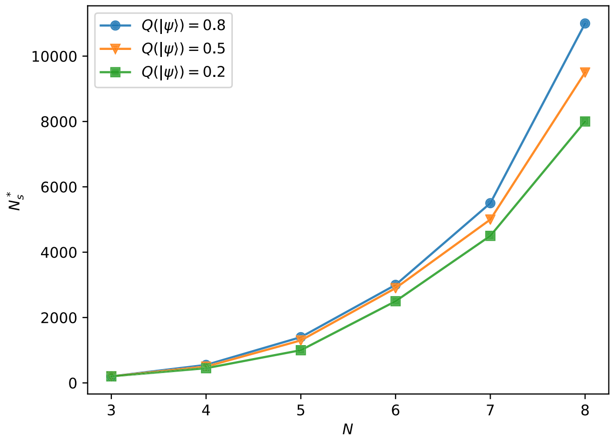


Fig. 7. The minimum number of training samples needed to achieve 0.95 fidelity for the quantum state generated by an MPS circuit with different entanglement entropies.

required to get a target fidelity (i.e. 0.95), especially for the states with large number of qubits.

4. Conclusion

In summary, we first use a powerful generative model (VAE) to reconstruct a relatively simple quantum state, i.e. the GHZ state. The simulation results show that VAE can reconstruct the GHZ state based on a small number of measurement outcomes with a high fidelity, i.e. 0.99. Due to the simple structure of the MPS circuit for the GHZ state, the number of measurement outcomes required to reconstruct the GHZ state scales approximately linearly with the number of qubits. As a supplement to current researches, we also study QST on a quantum state, i.e. the quantum state generated by a more complex MPS circuit, which has been widely used in quantum machine learning tasks. The simulation results show that we can reconstruct the quantum state with 0.95 fidelity, but it is difficult to achieve 0.99 fidelity. The number of measurement outcomes required for state reconstruction increases exponentially with the number of qubits, which shows the limitations of neural network tomography on complex quantum states. Our research provides a more comprehensive understanding of neural network tomography and put forward directions for future researches. In our future work, we will use Conditional Variational Autoencoder (CVAE) to learn the probability distribution of POVM outcomes of different quantum states.

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